

When Feedback Corrects the Wrong Thing

Cybernetics, Representational Feedback, and Reality Drift

Reality Drift Framework — Bridge Brief #1

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The Established Insight

Every adaptive system faces the same basic problem.

It must remain aligned with changing conditions.

An organism acts and encounters consequences.

A machine produces an output and measures the result.

An organization implements a policy and monitors performance.

An artificial intelligence system generates behavior and receives an evaluation signal.

The system then adjusts.

action → environmental response → feedback → correction → new action

Cybernetics provided one of the most powerful frameworks for understanding this process.

Its central insight is simple:

Systems remain aligned through feedback.

Without information about the effects of their behavior, systems cannot reliably correct themselves. Feedback makes adaptation, regulation, control, and error correction possible.

What Cybernetics Explains

Cybernetics studies communication and control within complex systems.

A thermostat measures temperature and adjusts heating.

An organism responds to changes in its environment.

A machine compares actual output with desired output.

An organization observes performance and modifies its procedures.

In each case, the system uses information about a difference:

current condition $\neq$ desired condition

That difference becomes an error signal.

The system acts to reduce the error.

The resulting loop is:

goal $\rightarrow$ action $\rightarrow$ feedback $\rightarrow$ comparison $\rightarrow$ correction

This structure explains how systems preserve stability while conditions change.

It also explains why systems without reliable feedback become brittle, unstable, or incapable of adaptation.

Cybernetics therefore gives Reality Drift a crucial starting point.

Alignment is not a fixed achievement.

It must be continually maintained through correction.

The Boundary

Many contemporary systems do not fail because feedback disappears.

The dashboard still updates.

The performance report still arrives.

The model still receives a reward signal.

The organization still collects user data.

The platform still measures behavior.

Communication, optimization, and adaptation remain active.

Yet the system becomes progressively less aligned with the reality it was created to navigate.

This creates a problem that cannot be explained by the simple absence of feedback.

The deeper question is:

What is the feedback actually correcting?

Large systems rarely interact directly with reality.

They interact with representations of reality:

- metrics
- dashboards
- reports
- benchmarks
- models
- forecasts
- narratives
- performance indicators

The feedback loop may remain fully operational while the representation supplying the correction signal becomes distorted, incomplete, or detached from the underlying conditions.

The Reality Drift Extension

Reality Drift shifts attention from the existence of feedback to the fidelity of the representation inside the feedback loop.

The ordinary cybernetic sequence appears to be:

reality → feedback → correction

In scaled symbolic systems, the actual sequence is often:

reality → representation → feedback signal → correction

This additional layer changes the problem.

The system does not correct itself against reality directly.

It corrects itself against a model, metric, report, benchmark, or interpretation of reality.

When that representation remains faithful, the loop can preserve alignment.

When the representation drifts, the loop may continue functioning while directing the system toward the wrong condition.

The system is not failing to correct.

It is correcting successfully against a degraded target.

Representational Feedback

Representational feedback is feedback derived from a proxy, model, or symbolic description rather than from direct contact with the underlying condition.

This is unavoidable in complex systems.

A hospital cannot manage itself through direct observation of every patient experience.

A corporation cannot inspect every consequence of every decision.

An AI system cannot receive a complete representation of human intention.

Institutions therefore rely on compressed signals.

The danger begins when the system treats the signal as though it were equivalent to the reality it represents.

The sequence becomes:

underlying condition → proxy signal → optimization → proxy improvement → weakened correspondence

Feedback remains active.

Correction remains active.

The system becomes increasingly responsive to the proxy and decreasingly answerable to the world beneath it.

The Bridge

Cybernetics and Reality Drift address different layers of alignment.

Cybernetics asks:

How does a system correct its behavior through feedback?

Reality Drift asks:

What happens when the representation supplying that feedback gradually loses correspondence with reality?

Cybernetics explains correction.

Reality Drift examines the integrity of the object being corrected against.

This produces a distinct failure mode:

accurate correction + inaccurate representation = systematic misalignment

The system may become faster, more adaptive, and more internally coherent while moving further from its actual purpose.

Feedback does not disappear.

It becomes enclosed within the representational environment of the system.

Where the Distinction Matters

Organizations

An organization monitors productivity through completed tasks.

Employees learn to maximize visible completion.

The reporting system shows improvement while useful work becomes harder to identify.

The feedback loop functions.

The proxy has replaced the outcome.

Artificial Intelligence

A model receives continuous optimization signals based on an evaluation function.

Performance against the function improves.

The relationship between the reward and the intended objective becomes less certain.

The model has not escaped feedback.

It has learned the feedback environment.

Digital Platforms

A platform measures engagement and adapts recommendations continuously.

The algorithm becomes highly responsive to user behavior.

The system successfully increases interaction while potentially weakening user well-being, informational quality, or cultural value.

The feedback is real.

The represented objective is incomplete.

Institutions

An agency monitors compliance, processing time, and procedural completion.

Reports remain current and accurate.

The institution becomes increasingly effective at satisfying administrative indicators while becoming less effective at serving its original mission.

The system continues correcting.

It corrects toward institutional representation rather than public reality.

Core Distinction

Cybernetics explains how systems remain aligned through feedback and correction.

Reality Drift examines how alignment can weaken when the representations inside those feedback loops become detached from the realities they were designed to track.

Feedback corrects the wrong thing when the loop remains functional but the error signal no longer refers reliably to the underlying condition.